

Hypothesis testing with multiple regression

Hypothesis testing with a categorical predictor

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

We want to know whether there is a relationship between species and bill length.

Can I translate this into null and alternative hypotheses about one or more model parameters?

$$H_0: \beta_1 = \beta_2 = 0$$

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$$H_0 : \beta_1 = \beta_2 = 0$$

$$H_A : \text{at least one of } \beta_1, \beta_2 \neq 0$$

Full and reduced models

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

$$H_0 : \beta_1 = \beta_2 = 0$$

$$H_A : \text{at least one of } \beta_1, \beta_2 \neq 0$$

Reduced model (H_0):

$$\text{bill length} = \beta_0 + \varepsilon$$

Full model (H_A):

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

What test did we use to test these hypotheses?

F test

Variability:

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$SSTotal = SSModel + SSE$$

Degrees of freedom:

$$n - 1 = (p - 1) + (n - p)$$

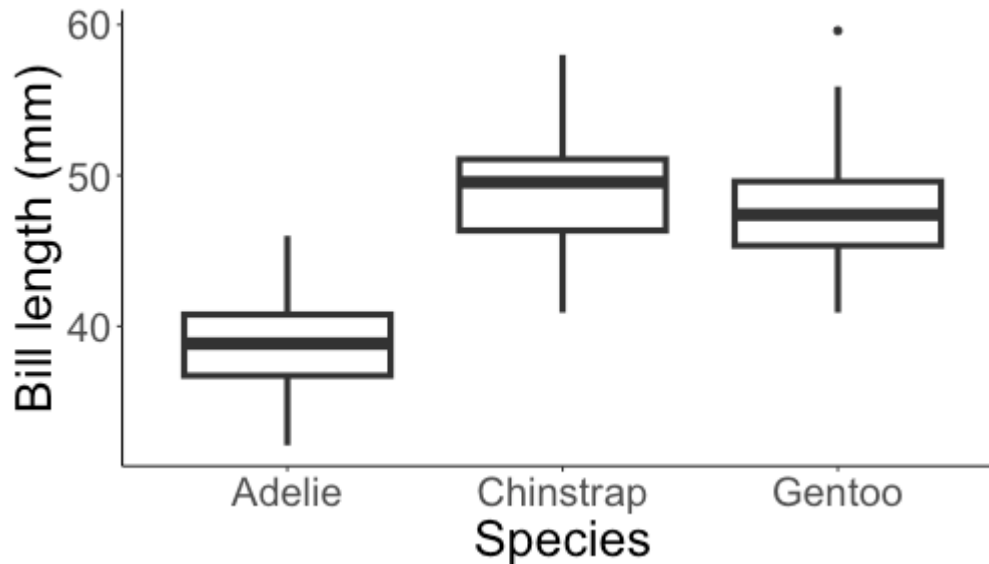
(n = number of observations, p = number of parameters in full model)

Test statistic:

$$F = \frac{\text{between group variance}}{\text{within group variance}} = \frac{SSModel/(p - 1)}{SSE/(n - p)}$$

Under H_0 , $F \sim F_{p-1, n-p}$

F test



Test statistic:

$$F = \frac{\text{between group variance}}{\text{within group variance}} = \frac{SSModel/(p-1)}{SSE/(n-p)} = 397.3$$

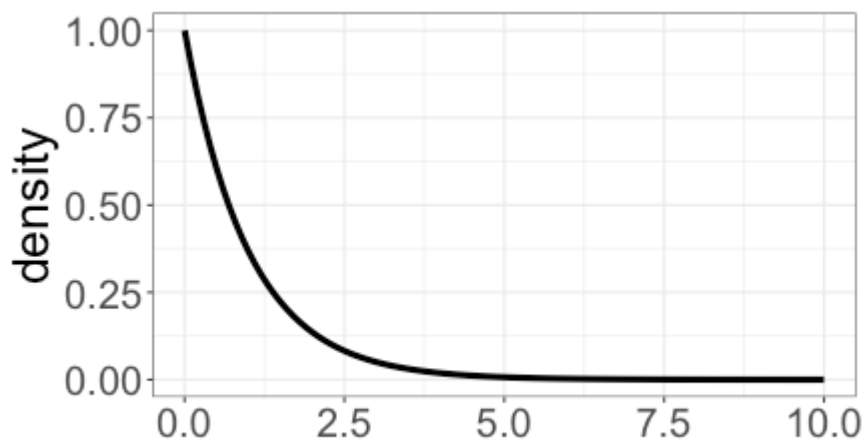
How do I calculate a p-value?

Calculating p-values

$$\text{If } H_0 \text{ is true, } F = \frac{SSModel/(p-1)}{SSE/(n-p)} \sim F_{p-1, n-p}$$

Degrees of freedom: $p-1 = 2, n-p = 330$

$F_{2, 330}$ distribution:



$$\text{p-value} = P(F_{2, 330} > 397.3) \approx 0$$

Summary so far

Our F test compares two *nested* models:

Reduced model (H_0):

$$\text{bill length} = \beta_0 + \varepsilon$$

Full model (H_A):

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

Test statistic: $F = \frac{SSModel/(p-1)}{SSE/(n-p)} \sim F_{p-1, n-p}$

What if I want to test other hypotheses?

Hypotheses for multiple regression

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \beta_3 \text{body mass} + \varepsilon$$

Research question: Is there any relationship between the predictors (species and body mass) and the response (bill length)?

How do I turn this question into hypotheses about one or more model parameters?

$$H_0: \beta_1 = \beta_2 = \beta_3 = 0$$

$$H_A: \text{at least one of } \beta_1, \beta_2, \beta_3 \neq 0$$

Hypotheses for multiple regression

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \beta_3 \text{body mass} + \varepsilon$$

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$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0 \quad H_A : \text{at least one of } \beta_1, \beta_2, \beta_3 \neq 0$$

What are the full and reduced models?

Full and reduced models

$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ $H_A : \text{at least one of } \beta_1, \beta_2, \beta_3 \neq 0$

Full model (H_A):

$$\begin{aligned} \text{bill length} = & \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ & + \beta_3 \text{body mass} + \varepsilon \end{aligned}$$

Reduced model (H_0):

$$\text{bill length} = \beta_0 + \varepsilon$$

Note that the reduced model is *nested* inside the full model.

Hypotheses for multiple regression

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \beta_3 \text{body mass} + \varepsilon$$

Research question: After accounting for species, is there a relationship between body mass and bill length?

How do I turn this question into hypotheses about one or more model parameters?

$$H_0: \beta_3 = 0$$

$$H_A: \beta_3 \neq 0$$

Hypotheses for multiple regression

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \beta_3 \text{body mass} + \varepsilon$$

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Reduced model (H_0):

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

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Hypotheses for multiple regression

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$$H_0: \beta_1 = \beta_2 = 0$$

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$$H_0 : \beta_1, \beta_2 = 0 \quad H_A : \text{at least one of } \beta_1, \beta_2 \neq 0$$

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Full and reduced models

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$$\begin{aligned} \text{bill length} = & \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ & + \beta_3 \text{body mass} + \varepsilon \end{aligned}$$

Reduced model (H_0):

$$\text{bill length} = \beta_0 + \beta_1 \text{body mass} + \varepsilon$$

Note that the reduced model is *nested* inside the full model.

Hypotheses for multiple regression

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \beta_3 \text{body mass} + \varepsilon$$

Research question: Is the slope on body mass different for different species?

Can I test this research question with this model?

Hypotheses for multiple regression

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \beta_3 \text{body mass} + \varepsilon$$

Research question: Is the slope on body mass different for different species?

Can I test this research question with this model?

No -- the model assumes that the slope is the same for each species!

How can I change the model to test this research question?

Adding interaction terms

$$\begin{aligned}\text{bill length} = & \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ & + \beta_3 \text{body mass} + \beta_4 \text{body mass} \cdot \text{IsChinstrap} \\ & + \beta_5 \text{body mass} \cdot \text{IsGentoo} + \varepsilon\end{aligned}$$

Research question: Is the slope on body mass different for different species?

How do I turn this question into hypotheses about one or more model parameters?

$$H_0: \beta_u = \beta_s = 0$$

$$H_A: \text{at least one of } \beta_u, \beta_s \neq 0$$

Adding interaction terms

$$\begin{aligned}\text{bill length} = & \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ & + \beta_3 \text{body mass} + \beta_4 \text{body mass} \cdot \text{IsChinstrap} \\ & + \beta_5 \text{body mass} \cdot \text{IsGentoo} + \varepsilon\end{aligned}$$

Research question: Is the slope on body mass different for different species?

How do I turn this question into hypotheses about one or more model parameters?

$$H_0 : \beta_4, \beta_5 = 0 \quad H_A : \text{at least one of } \beta_4, \beta_5 \neq 0$$

What are the full and reduced models?

Testing interaction terms

$$H_0 : \beta_4, \beta_5 = 0 \quad H_A : \text{at least one of } \beta_4, \beta_5 \neq 0$$

Full model (H_A):

$$\begin{aligned} \text{bill length} = & \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ & + \beta_3 \text{body mass} + \beta_4 \text{body mass} \cdot \text{IsChinstrap} \\ & + \beta_5 \text{body mass} \cdot \text{IsGentoo} + \varepsilon \end{aligned}$$

Reduced model (H_0):

$$\begin{aligned} \text{bill length} = & \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ & + \beta_3 \text{body mass} + \varepsilon \end{aligned}$$

Note that the reduced model is *nested* inside the full model.

Class activity

Work on the class activity (handout).

Class activity

Research question: Is the slope on Age the same for both boys and girls?

What model should the researchers use to test this research question?

$$\text{Height} = \beta_0 + \beta_1 \text{IsMale} + \beta_2 \text{Age} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Class activity

Research question: Is the slope on Age the same for both boys and girls?

What model should the researchers use to test this research question?

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

What are the full and reduced models that the researchers will compare?

$H_0: \beta_3 = 0 \rightarrow$ remove interaction term

Class activity

Full model:

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Reduced model:

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \varepsilon$$

Class activity

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

New research question: Is there *any* relationship between Age and Height, after accounting for gender?

What are the full and reduced models that the researchers will compare?

Class activity

New research question: Is there *any* relationship between Age and Height, after accounting for gender?

Full model:

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Reduced model:

$$\text{Height} = \beta_0 + \beta_1 \text{IsMale} + \varepsilon$$

What we've done so far

- + Specifying hypotheses for multiple regression models
- + Hypotheses imply full and reduced models

How do we compare full and reduced models with a hypothesis test?

When our reduced model is intercept-only

We've already done this for one categorical predictor!

Reduced model (H_0):

$$\text{bill length} = \beta_0 + \varepsilon \quad (\text{intercept-only model})$$

Full model (H_A):

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

Test statistic: $F = \frac{SSModel/(p-1)}{SSE/(n-p)} \sim F_{p-1, n-p}$

What if I want to compare different nested models?

Decomposing sums of squares

$$F = \frac{SSModel/(p-1)}{SSE/(n-p)}$$

- ✚ SSModel and SSE really refer to the *full* model ($\hat{y}_i$ are predictions from the *full* model, not the reduced model)
- ✚ So let's add subscripts to make that clear

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_{i,full} - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_{i,full})^2$$

$$SSTotal = SSModel_{full} + SSE_{full}$$

reduction in variance from fitting model

$$F = \frac{SSModel_{full}/(p-1)}{SSE_{full}/(n-p)} = \frac{(SSTotal - SSE_{full})/(p-1)}{SSE_{full}/(n-p)}$$

↑ amount of variability left after fitting model

Decomposing sums of squares

$$F = \frac{(SSTotal - SSE_{full}) / (p - 1)}{SSE_{full} / (n - p)}$$

We can decompose sums of squares for the reduced model too:

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_{i, reduced} - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_{i, reduced})^2$$

$$SSTotal = SSModel_{reduced} + SSE_{reduced}$$

For this test, the reduced model is

$$\text{bill length} = \beta_0 + \varepsilon$$

What are the predictions $\hat{y}_{i, reduced}$ for this intercept-only model?

$$\hat{y}_{i, reduced} = \hat{\beta}_0 = \bar{y}$$

Decomposing sums of squares

$$F = \frac{(SSTotal - SSE_{full}) / (p - 1)}{SSE_{full} / (n - p)}$$

We can decompose sums of squares for the reduced model too:

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\underbrace{\hat{y}_{i, reduced} - \bar{y}}_{\substack{\hookrightarrow 0 \text{ if } \hat{y} = \bar{y}}})^2 + \sum_{i=1}^n (y_i - \hat{y}_{i, reduced})^2$$
$$SSTotal = SSModel_{reduced} + SSE_{reduced}$$

Reduced model: bill length = $\beta_0 + \varepsilon$

Predictions: $\hat{y}_{i, reduced} = \bar{y}$

SS Total ↙
If $\hat{y}_{i, reduced} = \bar{y}$, what are $SSModel_{reduced}$ and $SSE_{reduced}$?

Decomposing sums of squares

$$F = \frac{(SSTotal - SSE_{full})/(p - 1)}{SSE_{full}/(n - p)}$$

We can decompose sums of squares for the reduced model too:

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_{i, reduced} - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_{i, reduced})^2$$

$$SSTotal = SSModel_{reduced} + SSE_{reduced}$$

Reduced model: bill length = $\beta_0 + \varepsilon$

Predictions: $\hat{y}_{i, reduced} = \bar{y}$

$$SSModel_{reduced} = 0 \quad SSE_{reduced} = SSTotal$$

Decomposing sums of squares

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_{i, \text{reduced}} - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_{i, \text{reduced}})^2$$

$$SSTotal = SSModel_{\text{reduced}} + SSE_{\text{reduced}}$$

Reduced model: bill length = $\beta_0 + \varepsilon$

Predictions: $\hat{y}_{i, \text{reduced}} = \bar{y}$

$$SSModel_{\text{reduced}} = 0 \quad SSE_{\text{reduced}} = SSTotal$$

$$\begin{aligned} F &= \frac{(SSTotal - SSE_{\text{full}})/(p - 1)}{SSE_{\text{full}}/(n - p)} \\ &= \frac{(SSE_{\text{reduced}} - SSE_{\text{full}})/(p - 1)}{SSE_{\text{full}}/(n - p)} \end{aligned}$$

Re-writing our F test

Our F test compares two *nested* models:

Reduced model (H_0):

$$\text{bill length} = \beta_0 + \varepsilon$$

Full model (H_A):

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

Test statistic:

$$F = \frac{(SSE_{\text{reduced}} - SSE_{\text{full}})/(p - 1)}{SSE_{\text{full}}/(n - p)} \sim F_{p-1, n-p}$$

- + $p - 1$ = difference in number of parameters between models
- + p = number of parameters in full model

Nested F test

To compare *any* two **nested** models:

Test statistic: $F = \frac{\frac{1}{\# \text{ parameters tested}} (SSE_{\text{reduced}} - SSE_{\text{full}})}{\frac{1}{n-p} SSE_{\text{full}}}$

how much better does our full model predict the response

how much variability is left after fitting full model

Distribution: $F_{\# \text{ parameters tested}, n-p}$

- + n = number of observations in data
- + p = number of parameters in full model

e.g.

$$H_0: \beta_1 = \beta_2 = 0$$

$$\# \text{ parameters tested} = 2$$